Mohamed Amine Mechlia Ingénieur Deep Learning

mechliamohamedamine@gmail.com +33 06 66 24 01 27 Paris, France Website

LinkedIn GitHub

Deep Learning Engineer

Machine Learning Specialist with extensive experience in applying advanced data science techniques, including deeplearning and computer vision, to solve complex business problems. With 3+ years of experience as a Deep Learning Specialist, I have applied advanced techniques in computer vision, embedded AI, and real-time image processing to both research and industry projects. Skilled in Python, C++, TensorFlow, PyTorch, and embedded systems, with proven expertise in 3D reconstruction, hardware optimization, and satellite/ drone vision applications.

EXPÉRIENCE PROFESSIONNELLE

R&D engineer computer vision Intern 03/2026 | croix, France
OVH cloud

• Developed an indoor localization and mapping system using Visual Odometry and Visual SLAM techniques.
• Designed real-time visual tracking algorithms for server localization in complex environments.
• Integrated ARKit-based augmented reality visualization and localization capabilities.
• Performed camera calibration, trajectory evaluation, system validation, and performance benchmarking. • Conducted testing and debugging under real-world deployment conditions.

Ingénieur Deep Learning Embarqué 03/2024 – 09/2025 | Tunisie
TELNET Space

- Conception et optimisation d'algorithmes de détection et de segmentation d'images satellitaires multispectrales pour déploiement embarqué.
- Développement d'un module de suppression de nuages (cloud removal) destiné à une démonstration en orbite.
- Création d'un accélérateur de vision embarqué pour edge computing satellite, conversion et optimisation de modèles PyTorch → ONNX, avec quantization et pruning pour NVIDIA Orin, réduisant la latence et l'utilisation mémoire.
- Déploiement d'une solution de détection de drones combinant traitement d'image et apprentissage automatique sur plateforme embarquée.
- Implémentation d'un système de communication CAN fiable entre l'ordinateur de bord (OBC) et l'unité de calcul embarquée pour assurer la transmission efficace des données.

Ingénieur Vision par Ordinateur Embarquée 11/2021 – 02/2024 | Irlande
Nurenda Technologies

• Developed SLAM-inspired localization solutions for GPS-denied navigation using visual perception and onboard sensors.
• Designed real-time Visual Odometry and pose estimation algorithms for autonomous drone navigation.
• Implemented embedded 3D reconstruction and scene understanding pipelines.
• Applied sensor fusion techniques combining camera, IMU, and navigation data. • Built synthetic datasets using Blender for localization, mapping, and computer vision training.
• Optimized computer vision algorithms for real-time embedded deployment.

FORMATION

Master Intelligence Artificielle et ses Applications en Vision et Robotique 2025 – en cours
Faculté des Sciences et Technologies de Nancy

Master en Électronique: Microélectronique et Instrumentation 2018 – 2021
Institut Supérieur d'Informatique et de Mathématiques de Monastir

COMPÉTENCES TECHNIQUES

- NVIDIA Jetson platform
- Quantization for NVIDIA Orin
- Vision accelerators for edge computing
- Drone detection systems
- Image segmentation
- SLAM, Odometry
- Système embarque
- ONNX model conversion
- Edge-to-cloud workflows
- Satellite image processing
- Deep learning
- Motion estimation
- Algorithm development
- Python, C/C++
- GPU/NPU/FPGA deployment
- Pruning techniques
- ADAS systems
- Object recognition and tracking
- Machine Learning
- 3D reconstruction (Nerf)
- Caméras (multispectral ,stereo Lidar...)

Projets

FPGA accelerator for Advanced Driver Assistance Systems (ADAS). 02/2020 – 03/2021
Laboratoire Micro-optoélectronique et nanostructures
- Development and optimization of an embedded neural network on FPGA for Advanced Driver Assistance Systems (ADAS).
- Design of a real-time pedestrian detection system capable of effectively preventing collisions.
- Performance optimization to reduce latency and maximize FPGA resource efficiency, ensuring fast and reliable processing for on-vehicle deployment.

Electronic Trap against the Olive Fruit Fly 02/2020 – 12/2020
IQFarm
- Development of a wireless connected monitoring system composed of automated traps for the identification and recognition of agricultural pests.
- Utilization of Raspberry Pi 0 to capture and transmit images to the AWS cloud.
- Implementation of cloud-based image processing and automatic recognition pipelines, featuring real-time notification of observations.
- Optimization of the edge-to-cloud data flow to ensure fast and reliable detection while minimizing the power consumption of embedded devices.

CERTIFICATION

- 3D Reconstruction - Multiple Viewpoints
- DeepLearning. AI TensorFlow Developer
- Machine Learning Concepts and Application of ML
- Natural Language Processing in TensorFlow
- IT Specialist - Python
- Scrum Fundamentals Certified (SFC)
- Sequences, Time Series and Prediction
- Convolutional Neural Networks